UNIT: 4

Joint distribution of discrete random variables :

1. Joint probability Function / Joint probability Distribution of Discrete Random Variables (DRV) :

Let X and Y be two discrete random variables. Then $f(x,y) = P(X=x, Y=y)$ is called joint probability function, if $f(x,y) \geq 0$ and $\sum_x \sum_y f(x,y) = 1$

Let X assumes the values $x_1, x_2, \cdots x_m$ and Y assumes the values $y_1, y_2, \cdots y_n$.
Then $f(x_i, y_j) = P(X=x_i, Y=y_j)$ represents the probability of the event that $X=x_i$ and $Y=y_j$
Where $i = 1, 2, \cdots m$ and $j = 1, 2, \cdots n$

2. Marginal probability Function :

The marginal probability function of random Variable X is given by $f_1(x_i) = P(X=x_i)$

x is constant, y will change

$$= \sum_{j=1}^{n} f(x_i, y_j) \qquad i = 1, 2, \cdots m$$

The marginal probability function of random Variable Y is given by $f_2(y_j) = P(Y=y_j) = \sum_{i=1}^{m} f(x_i, y_j)$

y will be constant & x will change

$$j = 1, 2, \cdots n$$

Joint probability Table :

No of hour

Scores in exam

X \ Y	y_1	y_2	. . .	y_n	Total.
x_1	(x_1, y_1)	(x_1, y_2)	. . .	(x_1, y_n)	$f_1(x)$
x_2	(x_2, y_1)	(x_2, y_2)	. . .	(x_2, y_n)	$f_1(x)$
⋮	⋮			⋮	⋮
x_m	(x_m, y_1)	(x_m, y_2)	. . .	(x_m, y_n)	$f_1(x)$
Total	$f_2(y_1)$	$f_2(y_2)$	. . .	$f_2(y_n)$	Grand Total $=1$

Marginal probability function of X

Marginal probability function of Y

- Marginal probability function of $X = x_i$ can be obtained by adding all entries in the row corresponding to x_i.
- Marginal probability function of $Y = y_j$ can be obtained by adding all the entries in the column X corresponding to y_j.
- Further $\sum_{i=1}^{m} f_1(x_i) = 1$, $\sum_{j=1}^{n} f_2(y_j) = 1$ or

$$\sum_{i=1}^{m} \sum_{j=1}^{n} f(x_i, y_j) = 1$$

3. Joint discribution function $F(x,y)$ of discrete random variables :

$$F(x,y) = P(X \le x, Y \le y)$$
$$= P(-\infty < X \le x, -\infty < Y \le y)$$
$$= \sum_{u \le x} \sum_{v \le y} f(u,v)$$

i.e $F(x,y)$ is the sum of all entries for which $x_i \le x$ and $y_i \le y$

4. Independent Variables :

If the events $X = x_i$ and $Y = y_j$ are independent events for all i and j, then X and Y are called independent random variables.

In this case, $f(x,y) = f_1(x) \cdot f_2(y)$

OR //

$$P(X=x, Y=y) = P(X=x) \cdot P(Y=y)$$

Q. Let X and Y be random variable having joint probablity function

$$f(x,y) = \begin{cases} cxy & , x = 1,2,3 , y = 1,2,3 \\ 0 & , \text{otherwise} \end{cases}$$

find : (i) c (ii) Marginal probability function of X and Y

(iii) $P(X=3)$ (iv) $P(X>1)$

(v) Check whether X and Y are independent

(vi) $P(X=3, Y=1)$ (vii) $P(X \geq 2)$

(viii) $P(X<2)$ (ix) $P(1 \le X \le 2, Y \geq 3)$

Sol^n: Joint probability table is

X \ Y	1	2	3	Total
1	c	2c	3c	6c
2	2c	4c	6c	12c
3	3c	6c	9c	18c
Total	6c	12c	18c	$\boxed{36c}$ → Grand total

(i) Since $f(x,y)$ is probability function

$$\therefore \sum_x \sum_y f(x,y) = 1 = \text{Grand total}$$

$$36c = 1$$

$$c = \frac{1}{36}$$

(ii) Marginal probability function of X is given by

$$f_1(x) = \begin{cases} 6c = \cancel{6} \times \frac{1}{36} = \frac{1}{6}, & x=1 \\ 12c = \cancel{6} \times \frac{\cancel{12}\,2}{\cancel{36}\,\cancel{6}\,3} = \frac{1}{3}, & x=2 \\ 18c = \frac{\cancel{18}}{\cancel{6}} \times \frac{1}{\cancel{36}_{\cancel{12}\,2}} = \frac{1}{2}, & x=3 \end{cases}$$

$$f_1(x) = \begin{cases} \frac{x}{6}, & x=1,2,3 \\ 0, & \text{otherwise} \end{cases}$$

Marginal probability function of Y is given by

$$f_2(y) = \begin{cases} 6c = 1/6, & y=1 \\ 12c = 1/3, & y=2 \\ 18c = 1/2, & y=3 \end{cases}$$

i.e $f_2(y) = \begin{cases} y/6 & , y = 1,2,3 \\ 0 & , \text{otherwise} \end{cases}$

(iii) $P(X=3) = 3c + 6c + 9c = 18c = 18 \cdot \frac{1}{36} = \frac{1}{2}$

(iv) $P(X>1) = (2c + 4c + 6c) + (3c + 6c + 9c)$

$= 12c + 18c$

$= 30c$

$= \frac{\cancel{30}}{5} \times \frac{1}{\cancel{36}\,6}$

$= \frac{5}{6}$

(v) $f_1(x) \cdot f_2(y) = \begin{cases} \frac{xy}{36} & , x = 1,2,3 \ , y = 1,2,3 \\ 0 & , \text{otherwise} \end{cases}$

$f_1(x) \cdot f_2(y) = f(x,y)$

i.e probability function is equal to product of function of x alone and y alone.

$\therefore$ X and Y are Independent random variables

(vi) $P(X=3, Y=1) = 3c = 3 \times \frac{1}{36} = \frac{1}{12}$

(vii) $P(X \geq 2) = (2c + 4c + 6c) + (3c + 6c + 9c)$

$= 12c + 18c$

$= 30c$

$= 30 \times \frac{1}{36}$

$= 5/6$

(viii) $P(Y<2) = c+2c+3c$

$= 6c$

$= 6 \times \frac{1}{36}$

$= \frac{1}{6}$

(ix) $P(1 \leq X \leq 2, Y \geq 3) = 3c + 6c = 9c = 9 \times \frac{1}{36} = \frac{1}{4}$

Joint distribution function / of continuous random variables :

1. Joint probability function / Joint probability distribution / Joint density function

Let X and Y be two continuous random variable Then $f(x,y)$ is called probability function or joint density function if

(i) $f(x,y) \geq 0$ (ii) $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dy\,dx = 1$

Note : (1) $P(a \leq X \leq b, c \leq Y \leq d) = \int_{x=a}^{b}\int_{y=c}^{d} f(x,y)\,dy\,dx$

(2) We can replace symbol $\leq$ by $<$

2. Joint distribution function :

The joint distribution function $F(x,y)$ of two continuous random variables is defined as

$$F(x,y) = P(X \le x, Y \le y)$$
$$= P(-\infty < X \le x, -\infty < Y \le y)$$
$$= \int_{u=-\infty}^{x} \int_{v=-\infty}^{y} f(u,v)\, du \cdot dv$$

Note : $\dfrac{\partial^2 F}{\partial x \cdot \partial y} = f(x,y)$

3. Marginal distribution function of X and Y :

Marginal distribution function of X is given by

$$F_1(X) = P(X \le x) = \int_{u=-\infty}^{x} \int_{v=-\infty}^{\infty} f(u,v)\, dv \cdot du$$

Marginal distribution function of Y is given by

$$F_2(Y) = P(Y \le y) = \int_{u=-\infty}^{\infty} \int_{v=-\infty}^{y} f(u,v)\, dv \cdot du$$

4. Marginal density function of X :

$$f_1(x) = \int_{v=-\infty}^{\infty} f(x,v)\, dv$$

Here, x will be constant
$y = v$ will be change

Marginal density function of Y :

$$f_2(y) = \int_{u=-\infty}^{\infty} f(u,y)\, du$$

Here y will be change
& $x = u$ will be change

Q. Let X and Y be continuous random variables having joint density function:

$$f(x,y) = \begin{cases} c(2x+y), & 0<x<1,\ 0<y<2 \\ 0, & \text{otherwise} \end{cases}$$

Find: (i) The constant c

(ii) The marginal density function of X

(iii) $P\left(X > \frac{1}{2}, Y < \frac{3}{2}\right)$

(iv) The marginal density function of Y

Sol^n: (i) Since $f(x,y)$ is joint density function we have $f(x,y) \geq 0$ and hence $c \geq 0$

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dy\,dx = 1$$

$$\int_0^1\int_0^2 c(2x+y)\,dy\,dx = 1$$

$$c\int_0^1 \left[2xy + \frac{y^2}{2}\right]_0^2 dx = 1$$

$$c\int_0^1 \left[2x2 + \frac{4}{2}\right] dx = 1$$

$$c\int_0^1 (4x+2)\,dx = 1$$

$$c\left(\frac{4x^2}{2} + 2x\right)_0^1 = 1$$

$$c(2+2) = 1$$

$$4c = 1$$

$$\boxed{c = 1/4}$$

(iii) $P\left(X > \frac{1}{2}, Y < \frac{3}{2}\right) = P\left(\frac{1}{2} < X < \infty, -\infty < Y < \frac{3}{2}\right)$

$$= \int_{1/2}^{\infty} \int_{-\infty}^{3/2} \frac{1}{4}(2x+y)\, dy\, dx$$

$$= \frac{1}{4} \int_{1/2}^{\infty} \left\{ 2xy + \frac{y^2}{2} \right\}_{-\infty}^{3/2} dx$$

$$= \frac{1}{4} \int_{1/2}^{\infty} \left\{ 3x + \frac{9}{4 \times 2} \right.$$

$$= \int_{1/2}^{1} \int_{0}^{3/2} \frac{1}{4}(2x+y)\, dy\, dx$$

$$= \frac{1}{4} \int_{1/2}^{1} \left(2xy + \frac{y^2}{2} \right)_{0}^{3/2} dx$$

$$= \frac{1}{4} \int_{1/2}^{1} \left(\frac{2x \times 3}{2} + \frac{9}{4 \times 2} \right) dx$$

$$= \frac{1}{4} \int_{1/2}^{1} \left(3x + \frac{9}{8} \right) dx$$

$$= \frac{1}{4} \left(\frac{3x^2}{2} + \frac{9}{8} x \right)_{1/2}^{1}$$

$$= \frac{1}{4} \left[\left(\frac{3}{2} + \frac{9}{8} \right) - \left(\frac{3}{2}\left(\frac{1}{4}\right) + \frac{9}{8} \times \frac{1}{2} \right) \right]$$

$$= \frac{1}{4} \left[\frac{21}{8} - \left(\frac{3}{8} + \frac{9}{16} \right) \right]$$

$$= \frac{1}{4}\left[\frac{21}{8} - \frac{15}{16}\right]$$

$$= \frac{1}{4}\left[\frac{42-15}{16}\right]$$

$$= \frac{1}{4}\left(\frac{27}{16}\right)$$

$$P\left(X > \frac{1}{2},\ Y < \frac{3}{2}\right) = \frac{27}{64}$$

(ii) Marginal density function of X :

$$f_1(x) = \int_{v=-\infty}^{\infty} f(x,v)\,dv$$

$$= \int_0^2 \frac{1}{4}(2x+v)\,dv$$

$$= \frac{1}{4}\left[2xv + \frac{v^2}{2}\right]_0^2$$

$$= \frac{1}{4}\left[\cancel{4}\ 2x+2\right]$$

$$= \frac{2x+1}{2}$$

$$f_1(x) = \begin{cases} \frac{2x+1}{2} & , 0<x<1 \\ 0 & , \text{otherwise} \end{cases}$$

(iv) The marginal density function of Y :

$$f_2(Y) = \int_{u=-\infty}^{\infty} f(u,y)\,du$$

$$= \int_0^1 \frac{1}{4}(2u+y)\,du$$

$$= \frac{1}{4}\left[u^2 + uy\right]_0^1$$

$$= \frac{1}{4}(1+y)$$

$$f_2(y) = \begin{cases} \frac{1}{4}(1+y) & , \quad 0 < y < 2 \\ 0 & , \quad \text{otherwise} \end{cases}$$

Conditional Expectation, Variance and moments

(A) For discrete random variable:

Conditional expectation of X given $Y=y$ is denoted and defined as

$$E(X \mid Y=y) = \sum x \cdot f(x \mid y)$$

Conditional variance of $Y=y$ is denoted and defined as

$$Var(X \mid Y=y) = E[(X-\mu_x)^2 \mid Y=y]$$
$$= \sum (x-\mu_x)^2 f(x \mid y)$$

Where,

$$\mu_x = E(X=Y=y)$$

The rth conditional moment of X given Y about any value a is defined as

$$E[(X-a)^r \mid Y=y] = \sum_x (x-a)^r f(x \mid y)$$

(B) For Continuous Random Variable:

Similarly conditional expectation, Conditional Variance and r^{th} conditional moment of X given Y can be obtained by replacing sumation by integration.

Q. Let X and Y be random variables having probability function.

$$f(x,y) = \begin{cases} c(x^2+y^2), & x=0,1,2 \;,\; y=0,1,2 \\ 0 & , \text{ otherwise} \end{cases}$$

Find : (i) c (ii) Marginal densities

(iii) Find conditional expectation of Y given X and X given Y.

Sol^n: The joint probability table is,

X \ Y	0	1	2		
0	0	c	4c	5c	Marginal probability function of X
1	c	2c	5c	8c	
2	4c	5c	8c	17c	
	5c	8c	17c	$\boxed{30c}$	

Marginal probability function of Y

(i) Grand total = 1

$30c = 1$

$$\boxed{c = \frac{1}{30}}$$

(ii) Marginal probability function of X :

$$f_1(x) = \begin{cases} 5c & , x=0 \\ 8c & , x=1 \\ 17c & , x=2. \end{cases}$$

$$f_1(x) = \begin{cases} (3x^2+5)c & , x=0,1,2 \\ 0 & , \text{otherwise} \end{cases}$$

$$\therefore f_1(x) = \begin{cases} \dfrac{3x^2+5}{30}, & x=0,1,2 \\ 0, & \text{otherwise} \end{cases}$$

Marginal probability function of Y:

$$f_2(Y) = \begin{cases} 5c, & Y=0 \\ 8c, & Y=1 \\ 17c, & Y=2 \end{cases}$$

$$\therefore f_2(Y) = \begin{cases} \dfrac{3Y^2+5}{30}, & y=0,1,2 \\ 0, & \text{otherwise} \end{cases}$$

(iii) Conditional probability function of X given Y is

$$f(x|Y) = \frac{f(x,Y)}{f_2(Y)} = \frac{\dfrac{x^2+Y^2}{30}}{\dfrac{3Y^2+5}{30}}$$

$$= \frac{x^2+Y^2}{3Y^2+5}, \quad x=0,1,2, \quad Y=0,1,2$$

Similarly,

$$f(Y|x) = \frac{f(x,Y)}{f_1(x)} = \frac{x^2+Y^2}{3x^2+5}, \quad x=0,1,2, \quad Y=0,1,2$$

$\therefore$ Conditional expectation of X given Y is

$$E[X|Y] = \sum_{x=0}^{2} x \cdot f(x|Y)$$

$$= \sum_{x=0}^{2} \frac{x^2+Y^2}{3Y^2+5}$$

$$= 0 + \frac{1+y^2}{3y^2+5} + 2\left(\frac{4+y^2}{3y^2+5}\right)$$

$$= \frac{1+y^2+8+2y^2}{3y^2+5}$$

$$E(X|Y) = \frac{9+3y^2}{5+3y^2}$$

Conditional Expectation of Y given X is,

$$E[Y|X] = \sum_{y=0}^{2} y \cdot f(y|x)$$

$$= \sum_{y=0}^{2} y \cdot \frac{x^2+y^2}{3x^2+5}$$

$$= 0 + \frac{x^2+1}{3x^2+5} + 2\left(\frac{x^2+4}{3x^2+5}\right)$$

$$= \frac{x^2+1+2x^2+8}{3x^2+5}$$

$$= \frac{3x^2+9}{3x^2+5}$$